

ABC Retail Solutions – Data Engineering and Business Intelligence Project

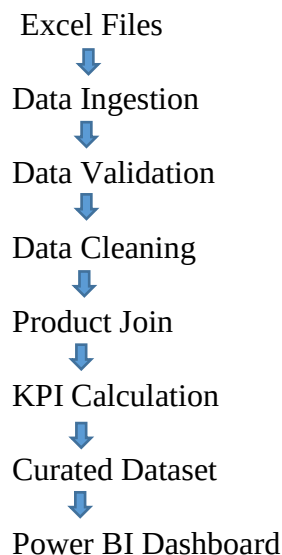
1. Project Overview

Problem Statement

ABC Retail Solutions operates across multiple cities through online and offline sales channels. Due to inconsistent retail transaction data, including duplicate records, missing values, invalid quantities, inconsistent product names, and varying date formats, the company faces challenges in generating accurate business insights.

The objective of this project is to design and implement an end-to-end data engineering pipeline that ingests, cleans, transforms, validates, and visualizes retail transaction data to support business intelligence and decision-making.

Architecture



3. Dataset Description

Product details

Contains product reference information.

Columns:

- Product ID
- Product Name
- Category
- Standard Product Price

Purpose:

Used as a dimension table to enrich retail transactions and fill missing prices.

retail_data1

Contains 4,243 retail transaction records.

Key Columns:

- Transaction ID
- Customer ID
- Product ID
- Product Name
- Category
- Quantity
- Price
- Discount
- Payment Method
- Payment Status
- City
- Purchase Location
- Email
- Phone
- Transaction Date

retail_data2

Contains 4,251 retail transaction records with the same structure as retail_data1.

4. Data Cleaning Strategy

Duplicate Removal

Removed duplicate transaction records using:

```
drop_duplicates()
```

Purpose:

Prevent double-counting of revenue and orders.

Missing Value Handling

Price

Missing prices were replaced using the standard product price from the product dimension table.

Quantity

Missing quantities were replaced with 1.

Discount

Missing discounts were replaced with 0.

City

Missing city values were replaced with "Unknown".

Product Name Standardization

Examples:

Laptop
LAPTOP
laptop

Converted into:

Laptop

using string formatting functions.

Category Standardization

Examples:

electronics
ELEC
Electronics

Converted into:

Electronics

5. PII Protection Strategy

Personally Identifiable Information (PII) was protected.

Email Masking

Example:

johnsmith@gmail.com

↓

jo****@gmail.com

Phone Number Hashing

SHA-256 hashing was applied to phone numbers.

Benefits:

- Data privacy

- Regulatory compliance
- Secure analytics

Power BI Dashboard Design

Page 1: Executive Summary

The Executive Summary page provides a high-level overview of the retail business performance. It enables stakeholders to quickly understand key business metrics and overall sales trends.

Visuals Included:

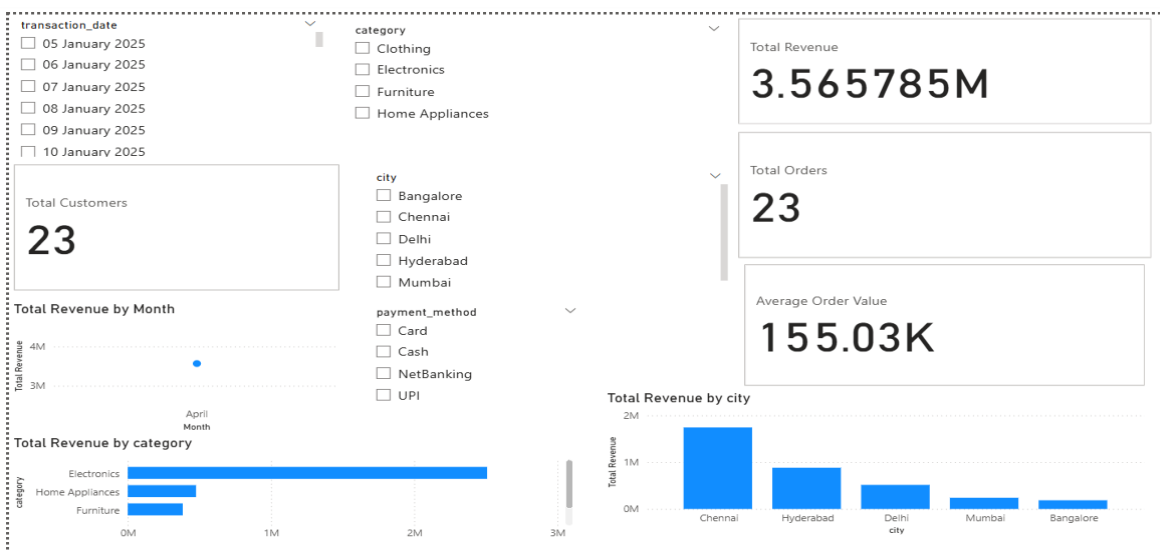
- Total Revenue
- Total Orders
- Total Customers
- Average Order Value
- Revenue by Category
- Revenue by City
- Revenue Trend by Month

Filters/Slicers:

- Transaction Date
- Category
- City
- Payment Method

Business Insights:

This page helps management monitor overall sales performance, customer activity, category contribution, and regional revenue distribution. Interactive slicers allow users to analyze business performance across different dimensions and time periods.



Page 2: Revenue Analysis

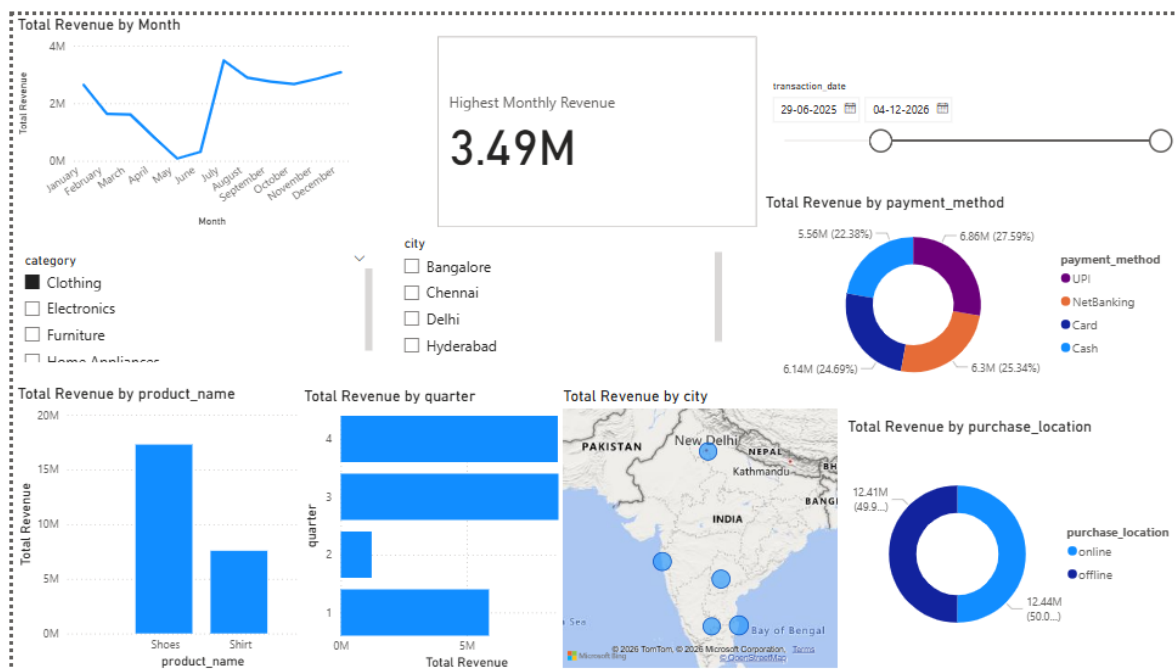
The Revenue Analysis page focuses on understanding revenue trends and sales channel performance.

Visuals Included:

- Revenue by Month
- Highest Monthly Revenue KPI
- Revenue by Quarter
- Revenue by Payment Method
- Revenue by Purchase Location (Online vs Offline)

Business Insights:

This page helps identify seasonal sales patterns, peak revenue periods, preferred payment methods, and the contribution of online and offline sales channels. These insights support strategic planning and revenue optimization.



Page 3: Product Performance

The Product Performance page evaluates the sales performance of products across the retail business.

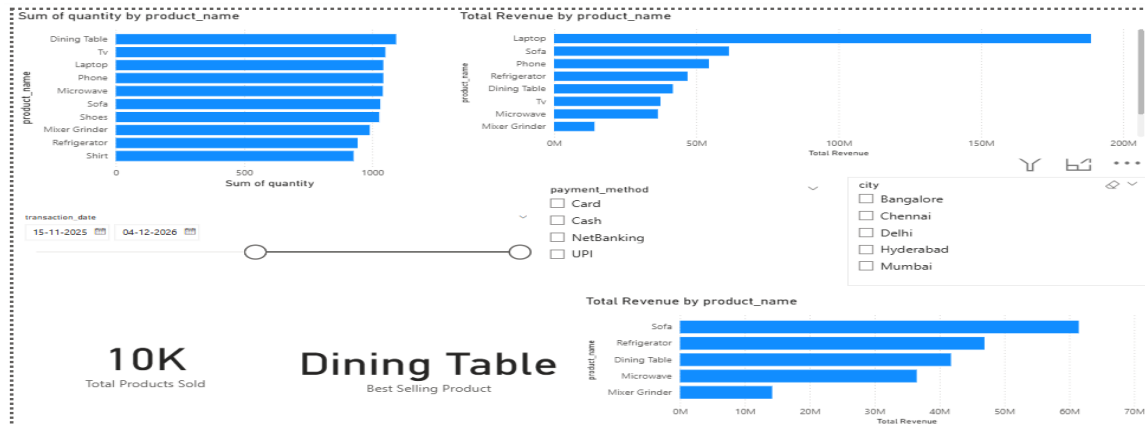
Visuals Included:

- Best Selling Product
- Total Products Sold
- Top Products by Revenue

- Quantity Sold by Product
- Revenue by Product

Business Insights:

This page identifies high-performing and low-performing products. It helps management optimize inventory planning, product promotions, and category management strategies.



Page 4: Category Trends

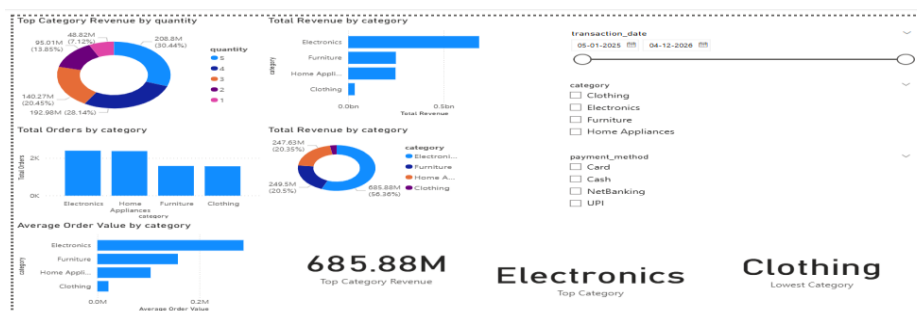
The Category Trends page analyzes the performance of different product categories.

Visuals Included:

- Revenue by Category
- Orders by Category
- Category-wise Average Order Value
- Category Contribution Percentage
- Top Category Revenue
- Top Category
- Lowest Category

Business Insights:

This page helps determine which product categories generate the highest revenue and customer demand. It also highlights underperforming categories, supporting product portfolio optimization and business decision-making.



Page 5: Regional Insights

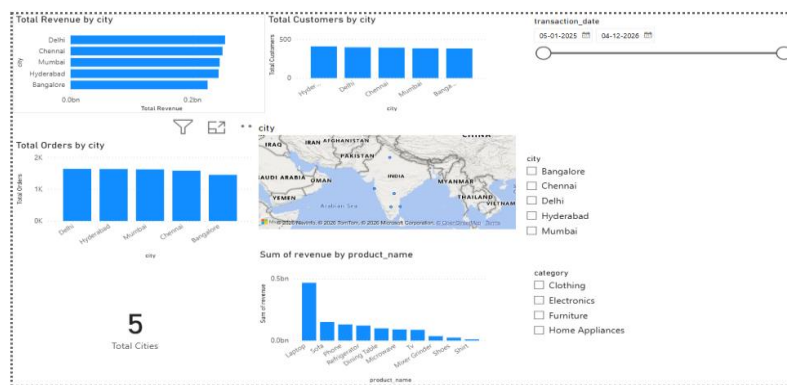
The Regional Insights page focuses on geographical sales performance across different cities.

Visuals Included:

- Revenue by City
- Total Customers by City
- Total Orders by City
- Revenue Distribution Map
- Total Cities KPI

Business Insights:

This page provides a regional view of business performance. Management can identify high-performing cities, understand customer concentration, and evaluate sales opportunities across locations. The map visualization enables easy geographical analysis of revenue distribution.



Page 6: Data Quality Dashboard

The Data Quality Dashboard demonstrates the data engineering and preprocessing activities performed during the project.

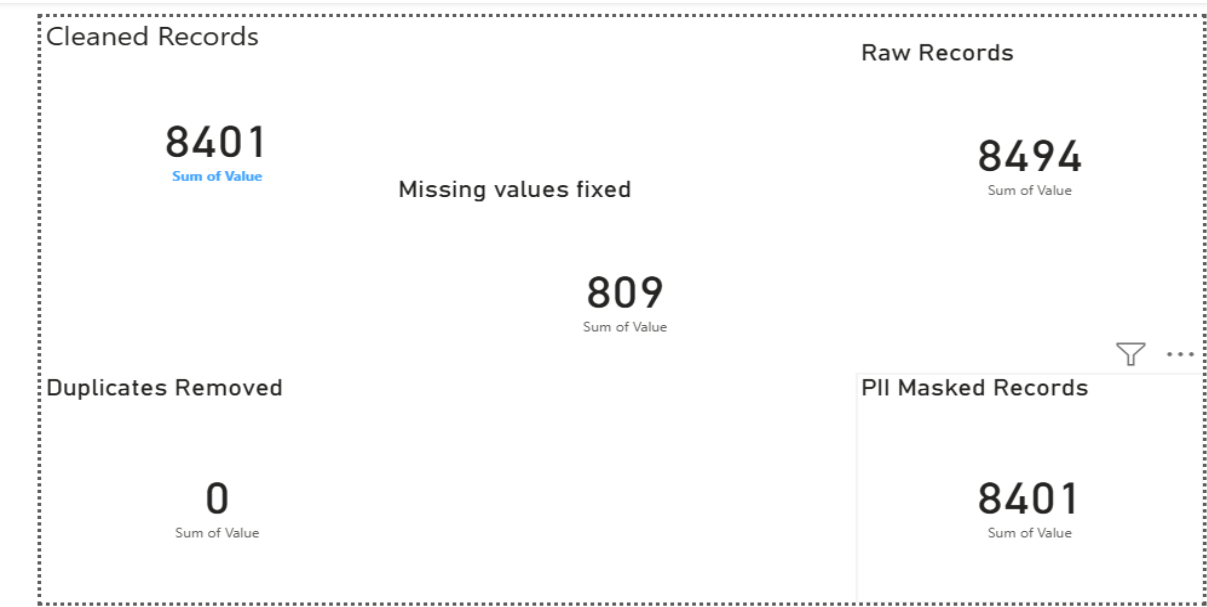
Visuals Included:

- Raw Records Count
- Clean Records Count
- Duplicate Records Removed
- Missing Values Fixed
- PII Masked Records
- Data Pipeline Flow Diagram

Business Insights:

This page highlights the quality improvement process applied to the raw retail datasets. It demonstrates duplicate removal, missing value handling, data standardization, and protection

of Personally Identifiable Information (PII). The page validates that the final dataset is reliable and ready for business intelligence reporting.



Results and Insights

Key Findings:

- Total Revenue Generated: ₹1,217,026,060
- Top Revenue Categories identified through category analysis.
- Top Performing Cities identified through regional analysis.
- Revenue trends visualized across time periods.
- Product performance insights generated for business decision-making.

Conclusion

The project successfully implemented an end-to-end retail data engineering pipeline. Data quality issues were addressed through validation and transformation processes, sensitive customer information was protected using masking and hashing techniques, and business-ready datasets were produced for Power BI reporting. The dashboard provides actionable insights into revenue, product performance, category trends, and regional sales performance.